

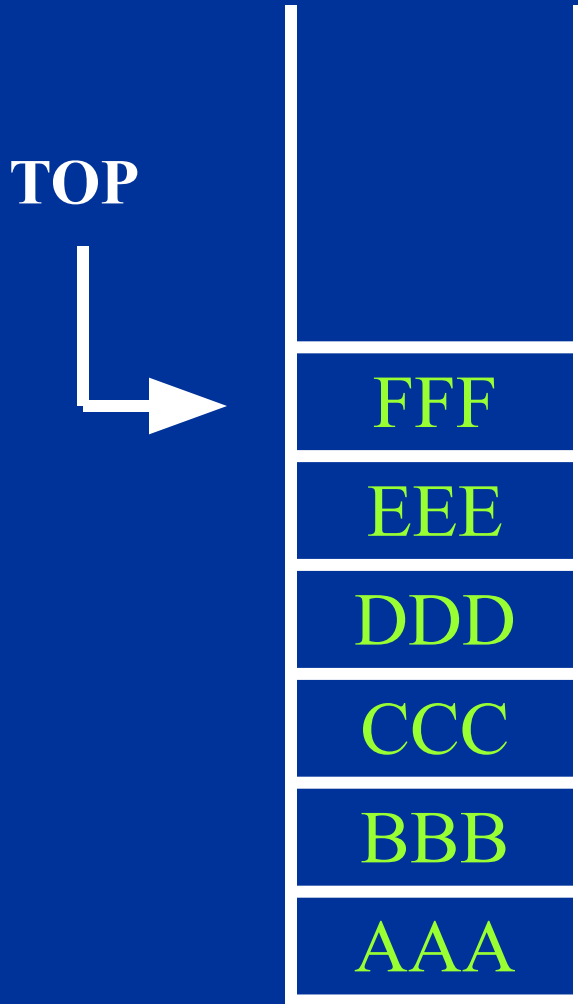
LECTURE – 5

What is a Stack?

A stack is a list of elements in which an element may be inserted or deleted only at one end, called the top of the stack. A stack is a LIFO system.

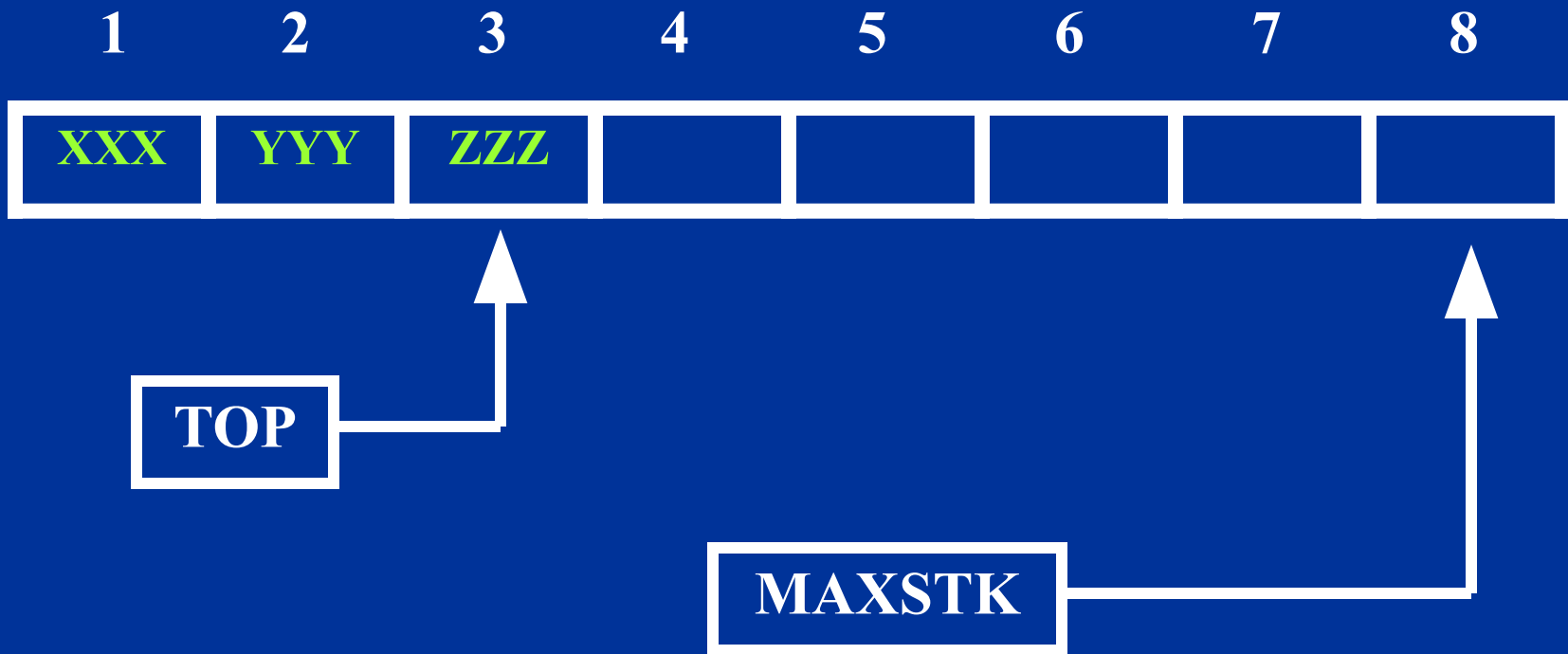
Special terminology is used for two basic Operations associated with stacks.

1. “Push” is the term used to insert an element into a stack
2. “Pop” is the term used to delete an element form a stack.



EEE cannot be deleted before **FFF** is deleted, **DDD** cannot be deleted before **EEE** and **FFF** are deleted and so on.

STACK DIAGRAM



Algorithm-1

PUSH (STACK, TOP, MAXSTK, ITEM)

Inserting an ITEM onto a stack

1. [stack already filled]

 If $TOP = MAXSTK$, then: Print: OVERFLOW, and return

2. Set $TOP := TOP + 1$

3. Set $STACK[TOP] := ITEM$

4. Return

Algorithm-2

POP (STACK, TOP, ITEM)

Deleting the top element of stack

1. [stack has no item to be removed]

 If $TOP = 0$, then: Print: UNDERFLOW, and return

2. Set $ITEM := STACK[TOP]$

3. Set $TOP := TOP - 1$

4. Return

QUICK SORT

Quick sort is a well-known sorting algorithm developed by **C. A. R Hoare** that makes comparisons to sort n items. Typically, quick sort is significantly faster in practice than other algorithms,

Suppose A is the following list of 12 numbers:

1	2	3	4	5	6	7	8	9	10	11	12
44	33	11	55	77	90	40	60	99	22	88	66

The “reduction step” of the quick sort algorithm finds the final position of one of the numbers.

In this illustration, we use the first number 44. beginning with last number 66, scan the list from right to left, comparing each number with 44 and stopping at the first number less then 44, then interchange the numbers.

1	2	3	4	5	6	7	8	9	10	11	12
22	33	11	55	77	90	40	60	99	44	88	66

beginning with 22, next scan the list in the opposite direction, from left to right, comparing each number with 44 and stopping at the first number greater than 44, then interchange the numbers.

1	2	3	4	5	6	7	8	9	10	11	12
22	33	11	44	77	90	40	60	99	55	88	66

beginning with 55, now scan the list in the original direction, from right to left, until meeting the first number less than 44, then interchange the numbers.

1	2	3	4	5	6	7	8	9	10	11	12
22	33	11	40	77	90	44	60	99	55	88	66

beginning with the 40, now scan the list from left to right, then interchange the numbers.

1	2	3	4	5	6	7	8	9	10	11	12
22	33	11	40	44	90	77	60	99	55	88	66

beginning with the 77, scan the list from right to left seeking a number less than 44. we do not meet such a number before meeting 44. This means all numbers have been scanned and compared with 44.

1	2	3	4	5	6	7	8	9	10	11	12
22	33	11	40	44	90	77	60	99	55	88	66

First Sublist Second Sublist

Procedure-1

QUICK (A, N, BEG, END, LOC)

1. Set $LEFT := BEG$, $RIGHT := END$, $LOC := BEG$
2. [scan from right to left]
 - a) repeat while $A[LOC] \leq A[RIGHT]$ and $LOC \neq RIGHT$
 $RIGHT := RIGHT - 1$
[end of loop]
 - b) if $LOC = RIGHT$, then: Return
 - c) if $A[LOC] > A[RIGHT]$, then:
 - (i) $TEMP := A[LOC]$, $A[LOC] := A[RIGHT]$
 $A[RIGHT] := TEMP$
 - (ii) set $LOC := RIGHT$
 - (iii) go to step 3

1	2	3	4	5	6	7	8	9	10	11	12
44	33	11	55	77	90	40	60	99	22	88	66

3. [scan from left to right]

a) repeat while $A[LEFT] \leq A[LOC]$ and $LEFT \neq LOC$

$LEFT := LEFT + 1$

[end of loop]

b) if $LOC = LEFT$, then: Return

c) if $A[LEFT] > A[LOC]$, then:

(i) $TEMP := A[LOC]$, $A[LOC] := A[LEFT]$

$A[LEFT] := TEMP$

(ii) set $LOC := LEFT$

(iii) go to step 2

[end of if structure]

1	2	3	4	5	6	7	8	9	10	11	12
22	33	11	55	77	90	40	60	99	44	88	66